That's a great idea. Students enjoy assignments much more when they can relate them to real life.

Assignment 3: Searching Algorithms in Real-Life Scenarios

Topics Covered: Linear Search & Binary Search

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1. Netflix Movie Search (Linear Search)

Scenario:

You open Netflix and want to find a movie in your "Watch Later" list. The app checks each movie one by one until it finds the movie.

Task:

Store movie IDs in an array and search for a specific movie ID using **Linear Search**.

Question:
Write a Java program to check whether the movie ID entered by the user exists in the watchlist.

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2. School Attendance Verification (Linear Search)

Scenario:

A teacher has a list of students present today. She checks the attendance sheet one by one to see if a student is present.

Task:

Store roll numbers in an array and search for a particular roll number using **Linear Search**.

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3. Parking Slot Search (Linear Search)

Scenario:

A security guard wants to know whether a car is parked in a parking area. He checks each parking slot one by one.

Task:

Store occupied parking slot numbers in an array and search for a specific slot number.

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4. Contact Search in Old Mobile Phone (Linear Search)

Scenario:

In an old mobile phone, contacts are searched one by one until the desired contact is found.

Task:

Store 20 contact numbers and search for a particular contact number.

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5. IPL Player Jersey Search (Linear Search)

Scenario:

A coach wants to check whether a jersey number belongs to a player in the team.

Task:

Store jersey numbers and search for a specific jersey number.

Binary Search Scenarios

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6. Dictionary Word Search (Binary Search)

Scenario:

When you search a word in a dictionary, you don't start from page 1. You open somewhere in the middle and keep reducing the search area.

Task:

Store sorted word IDs and search for a word ID using **Binary Search**.

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7. Finding a Book in a Library (Binary Search)

Scenario:

Books are arranged according to accession numbers. The librarian directly goes to the middle section and narrows the search.

Task:

Store sorted book IDs and search for a particular book ID using Binary Search.

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8. Searching a Student Result (Binary Search)

Scenario:

A school stores student IDs in sorted order. To find a student's result quickly, the system uses Binary Search.

Task:

Search a student ID in a sorted array.

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9. Train Seat Verification (Binary Search)

Scenario:

Railway seat numbers are stored in sorted order. The reservation system quickly checks whether a seat exists.

Task:

Store seat numbers in sorted order and search for a seat number using Binary Search.

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10. Online Shopping Product Search (Binary Search)

Scenario:

An e-commerce company stores product IDs in sorted order to search products faster.

Task:

Search a product ID using Binary Search.